

CHEM 3PC3

Assignment 1

Due date: Oct/23/2026 12:00 PM

Instructions

- Submit a single Jupyter notebook (.ipynb) containing your code, plots, and written answers. Write your **names, student IDs, and emails** at the top of the notebook. You can use Google Colab.
- Analytical derivations can be typed in Markdown cells (using $\text{\LaTeX}$) or handwritten, scanned, and attached. You can upload a photo of your mathematical derivation to an LLM and ask it for $\text{\LaTeX}$ code, which you can paste into your .ipynb file.
- Load **all** the libraries you need in the **first code cell** of your notebook. Importing libraries in any other cell will cost you marks. **[5 marks]**
- Every plot must label both axes with the variable *and its units*. Marks will be deducted for unlabelled axes.
- Functions must work on whole NumPy arrays: when a function is given a grid as input, its output must have the same size as that grid.
- In your .ipynb, label each section of your notebook with its question number (e.g. Q1.b).

Constants. Use $R = 0.083\,145\,\text{L bar/K/mol}$ throughout.

Question 1: The van der Waals equation of state

The van der Waals equation of state is

$$P(V, T) = \frac{RT}{V - b} - \frac{a}{V^2}, \quad (1)$$

where P is the pressure, T the temperature, V the molar volume, and a and b are the cohesion and excluded-volume constants of the gas. For nitrogen use,

$$a = 1.381\,69\,\text{bar L}^2/\text{mol}^2, \quad b = 0.0387\,\text{L/mol}.$$

These values are found by fitting the properties of the gas, and the van der Waals equation is normally used only to model the gas. In this assignment we will also use it to describe the liquid.

Q1.a van der Waals equation of state

[5 marks]

1. Write a Python function named `P_vdW(V, T)` that returns the pressure from Eq. (1).
2. Build a uniform grid of molar volumes with a total of 10.000 points from $V_{\min} = 0.045\,\text{L/mol}$ to $V_{\max} = 1\,\text{L/mol}$.
3. Plot P versus V for the isotherm $T = 77.25\,\text{K}$ (the normal boiling point of nitrogen).

Q1.b: Numerical differentiation**[10 marks]**

From the fundamental equation of thermodynamics, we can write the equality

$$\int_{P_1}^{P_2} V(P) dP = \int_{V_1}^{V_2} V \left(\frac{\partial P}{\partial V} \right)_T dV. \quad (2)$$

To evaluate the right-hand side, we need the partial derivative of the state function P with respect to V . One numerical approximation of a derivative is the **three-point forward difference formula**,

$$\frac{\partial f}{\partial x} \approx \frac{-3f(x) + 4f(x+h) - f(x+2h)}{2h}, \quad (3)$$

where f is the function you want to differentiate, x is the variable, and h is the step size.

1. Derive $\left(\frac{\partial P}{\partial V}\right)_T$ analytically for the van der Waals equation of state, Eq. 1. Show your work.
2. Write a Python function that evaluates your analytical derivative from 1. Name your function `d_P_vdW(V, T)`
3. Write a Python function that computes $\left(\frac{\partial P}{\partial V}\right)_T$ with the three-point forward difference formula, using $h = 1 \times 10^{-5}$ L/mol; name your function `d_P_vdW_3pf(V, T, h)`
4. Evaluate both functions, analytical and numerical, on the grid from Q1.a at $T = 77.25$ K, and compute the Frobenius norm of the difference between them,

$$\|\mathbf{d}_{\text{numerical}} - \mathbf{d}_{\text{analytical}}\|_F.$$

Comment on the result. Name your function `error_f`.

Q1.c: Numerical integration**[10 marks]**

In class, we saw that an integral can be approximated by a weighted sum of the integrand. For a uniform grid with spacing h and N grid points, the **composite three-point trapezoidal rule** is

$$\int_a^b f(x) dx \approx \frac{h}{4} \sum_{i=0}^{N-3} \left(f(x_i) + 2f(x_{i+1}) + f(x_{i+2}) \right). \quad (4)$$

1. Write a Python function for the three-point trapezoidal rule. Name your function `trap_3p_int`.
2. Evaluate the integral

$$I(V_1, V_2, T) = \int_{V_1}^{V_2} V \left(\frac{\partial P}{\partial V} \right)_T dV$$

analytically for the van der Waals equation of state. Show your work. *Hint:* write $\frac{V}{(V-b)^2} = \frac{1}{V-b} + \frac{b}{(V-b)^2}$.

3. Write a Python function that evaluates your analytical $I(V_1, V_2, T)$ for any V_1 , V_2 , and T . Name your function `analytical_int`.
4. Compute $I(V_1, V_2, T)$ with both methods for $V_1 = V_{\min}$, $V_2 = V_{\max}$, and $T = 77.25$ K, and report the absolute and relative errors of the numerical result.
5. Repeat the numerical integration with $N = 100$, 1000, and 10 000 grid points. In a plot, show how the error changes as N increases.

Question 2: Chemical potential and liquid–vapour coexistence

In this question you will use the van der Waals isotherm from Question 1 ($T = 77.25$ K, same values of $a = 1.38169$ bar L²/mol² and $b = 0.0387$ L/mol) to model the liquid–vapour phase transition of nitrogen.

Below the critical temperature, a horizontal line $P = P^\circ$ can cross the isotherm at three molar volumes. The smallest corresponds to a liquid-like state, and the largest to a gas-like state. The middle point lies on the unphysical branch where $(\frac{\partial P}{\partial V})_T > 0$. Two states can coexist in equilibrium only if they have the same pressure *and* the same chemical potential (molar Gibbs free energy).

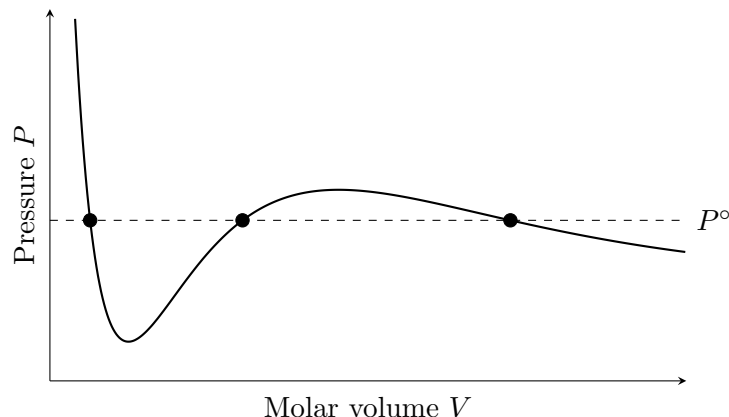


Figure 1: Schematic van der Waals isotherm below the critical temperature.

From the fundamental equation of thermodynamics,

$$dG = -S dT + V dP \implies d\mu = V dP \quad (\text{constant } T). \quad (5)$$

Since V is our independent variable, we can use Eq. (2) to write

$$\mu(V) = \mu(V_0) + \int_{V_0}^V V' \left(\frac{\partial P}{\partial V'} \right)_T dV'. \quad (6)$$

For this question, use a **logarithmically spaced** grid of molar volumes from $V_{\min} = 0.045$ L/mol to $V_{\max} = 50$ L/mol with 10 000 points (see `numpy.geomspace`). The gas branch at low pressure has a very large molar volume, so the grid from Question 1 does not reach it.

Q2.a: States with the same pressure

[5 marks]

1. Plot the isotherm P versus V on the new grid, using a logarithmic V axis. Add a horizontal line at $P^\circ = 1$ bar.
2. Find the first and third points (in order of increasing V) where $P = P^\circ$. To do this, find neighbouring grid points with $P_1 > P^\circ$ and $P_2 < P^\circ$ (or vice versa). Then estimate the crossing by linear interpolation, assuming P varies linearly between the two points:

$$P(V) \approx P_1 + \frac{P_2 - P_1}{V_2 - V_1} (V - V_1) \implies V^\circ = V_1 + (P^\circ - P_1) \frac{V_2 - V_1}{P_2 - P_1}. \quad (7)$$

Write a Python function that returns all such crossing volumes for any value of P° .

Q2.b: An expression for the chemical potential [5 marks]

Use your analytical integral from Q1.c to write $\mu(V) - \mu(V_0)$ for the van der Waals equation of state as an explicit function of V , V_0 , T , a , and b .

Q2.c: Evaluating the chemical potential [10 marks]

Set $\mu(V_0) = 0$, where V_0 is the smallest molar volume in your grid.

1. Evaluate and plot $\mu(V)$ on the grid.
2. Plot μ versus P , using the grid values of V as a parameter. Describe the shape of the curve and identify the liquid, gas, and unphysical branches.
3. Evaluate μ at the two $P^\circ = 1$ bar points from Q2.a. Do they have the same chemical potential? Which state is stable at 1 bar and 77.25 K?

Q2.d: Stability [5 marks]

For the pairs of points with $P = 0.25$ bar and with $P = 5$ bar, which point of each pair gives the stable equilibrium state? Justify your answer using the chemical potential.

Question 3: Adiabatic compression of carbon dioxide

One mole of carbon dioxide gas, initially at $T_1 = 25^\circ\text{C}$ and 1.00 bar, is compressed reversibly and adiabatically to *one tenth of its initial volume*, $V_2 = V_1/10$. Assume the gas is ideal, and use the empirical heat capacity

$$C_P(T) = (5.457 + 0.001045 T + 12100 T^{-2}) R, \quad (8)$$

where T is in kelvin. For an ideal gas, $C_V(T) = C_P(T) - R$.

Q3.a: Setting up the equation [5 marks]

For a reversible adiabatic process, $dU = -P dV$, and for an ideal gas $dU = C_V(T) dT$.

1. Show that the final temperature T_2 satisfies

$$\int_{T_1}^{T_2} \frac{C_V(T)}{T} dT = R \ln\left(\frac{V_1}{V_2}\right). \quad (9)$$

Hint: use $PV = RT$ (1 mol).

2. Evaluate the integral analytically, and rewrite the result as a nonlinear equation $F(T_2) = 0$.

Total: 60 marks